

Curriculum Vitae
Evan J. Morrison

Howard Hughes Medical Institute Gilliam Fellow

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(720)-270-5970

Education

- 2026 – HHMI Gilliam Postdoctoral Research Fellow
Stowers Institute for Medical Research, Kansas City, MO
Project: The signalosome regulates the quality control of defective large ribosomal subunit proteins in human cells
Advisor: Dr. Kamena Kostova
- 2018 – 2025 Ph.D. in Molecular Biology & Biochemistry
University of Colorado Anschutz Medical Campus, Aurora, CO
Thesis Project: Discovery and characterization of a MetAP-dependent N-degron pathway
Advisor: Dr. Olivia S. Rissland.
- 2011 – 2016 B.S. in Chemistry & B.A. in Philosophy
Metropolitan State University of Denver, Denver, CO
Research Project: Determination of Higher Order Structures in the Cap-Independent Translation Enhancer of the Blackcurrant Reversion 2 Virus.
Advisor: Dr. Megan Filbin-Wong.

Academic experience

- 09/2017 – 08/2018 Clinical Research Coordinator
University of Colorado Cancer Center
Job Duties: Investigate research protocols to determine subject eligibility for participation in clinical research trials. Obtain informed consent and conduct research visits with patients (take vitals, perform EKGs, examine critical lab values, and document and grade adverse events (CTCAE v.4.0)) to ensure patient safety. Process and ship biospecimens for laboratory analysis.
- 09/2016 – 09/2017 Laboratory Researcher
National Jewish Health
Center for Genes, Environment and Health
Research Project: Hepatic expression of chemokine ligand 20 is modulated by miR-590.
Advisor: Dr. Johanna K. DiStefano
- 06/2016 – 08/2016 RNA Bioscience Initiative Intern (REU)
University of Colorado School of Medicine
Department of Biochemistry & Molecular Genetics
Research Project: High throughput determination of higher order structures in the 3'UTR of a viral long noncoding RNA.
Advisor: Dr. Jeffrey Kieft.
- 06/2015 – 06/2016 Student Research Assistant

Metropolitan State University of Denver

Department of Chemistry

Research Project: Determination of higher order structures in the cap-independent translation enhancer of the blackcurrant reversion virus.

Advisor: Dr. Megan Filbin-Wong

Research skills

Laboratory techniques	Knowledge of biochemistry, molecular biology, and genetics experiments utilizing: gel electrophoresis, capillary electrophoresis, DNA purification, primer and G-block design, molecular cloning, transfection, transduction, RNA purification, <i>in vitro</i> transcription, <i>in vitro</i> translation, RT-qPCR, siRNA, miRNA, SHAPE and chemical probing, Western blot, ribosome fractionation and purification, immunoprecipitation, luciferase assays, LC-MS/MS, CRISPR KO, CRISPRi, CRISPRa, and mammalian cell culture.
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Grant support

09/2022 –	HHMI Gilliam Fellowship-GT14919 (PI: Morrison) Howard Hughes Medical Institute A tripeptide motif induces rapid protein decay in human cells
06/2021 – 08/2022	NSF GRFP-2020300571 (PI: Morrison) National Science Foundation Primary protein sequence regulates gene expression.
05/2020 – 04/2021	NIH T32-GM136444 (PI: Morrison) National Institute of Health/National Institute of General Medical Sciences Amino acid concentration influences ORF dependent mRNA stability

Publications

1. Leti F., **Morrison E.**, DiStefano JK. *Long noncoding RNAs in the pathogenesis of diabetic kidney disease: implications for novel therapeutic strategies*. Personalized Medicine. 2017,14(3): 271-8. DOI:10.2217/pme2016-0107.
2. **Morrison EJ.**, Champagne DP., Dzieciatkowska M., Nemkov T., Zimring JC., Hansen KC., Guan F., Huffman DM., Santambrogio L., and D'Alessandro A. *Parabiosis incompletely reverses aging-induced metabolic changes and oxidant stress in mouse red blood cells*. Nutrients. 2019, 11(6). DOI:10.3390/nu11061337.
3. Bertolone L., Kalani Roy M., Hay A., **Morrison EJ.**, Stefanoni D., Fu X., Kanas T., Gladwin M., Kleinman S., Dumont LJ., Stone M., Nemkov T., Busch M., Zimring JC., D'Alessandro A. *Impact of taurine on red blood cell metabolism and implications to blood storage*. Transfusion. 2020. DOI: 10.1111/trf.15810.
4. Bertolone L., Shin HKH., Stefanoni D., Baek JH., Gao Y., **Morrison EJ.**, Nemkov T., Thomas T., Francis RO., Hod EA., Zimring JC., Yoshida T., Karafin M., Schwartz J., Hudson KE., Spitalnik SL., Buehler

PW., D'Alessandro A. *ZOOMICS: comparative metabolomics of red blood cells from old world monkeys and humans*. *Frontiers in Physiology*. 2020. DOI: 10.3389/fphys.2020.593841.

5. D'Alessandro A., Hay A., Dzieciatkowska M., Brown BC., **Morrison EJ.**, Hansen KC., Zimring JC. *Protein-L-isoaspartate O-methyltransferase is required for in vivo control of oxidative damage in red blood cells*. *Haematologica*. 2020. DOI: 10.3324/haemtaol.2020266676.
6. Thomas T., Cendali F., Fu X., Gamboni F., **Morrison EJ.**, Beirne J., Nemkov T., Antonelou MH., Kriebardis A., Welsby I., Hay A., Dziewulska KH., Busch MP., Kleinman S., Buehler PW., Spitalnik SL., Zimring JC., D'Alessandro A. *Fatty acid desaturase activity in mature red blood cells and implications for blood storage quality*. *Transfusion* 2021. DOI: 10.1111/trf.16402.
7. Barrington CL., Galindo G., Koch AL., Horton E., **Morrison EJ.**, Tisa S., Stasevich TJ., Rissland OS. *Synonymous codon usage regulates translation initiation*. *Cell Reports* 2023. DOI: 10.1016/j.celrep.2023.113413
8. **Morrison EJ.***, Rao DM*, Horton ER*, Barrington CL*, Briney CA*, Henriksen JC*, Martinez-Seidel F*, Sterling EM*, Provencio ED., Harris M., Allen E., Hesselberth J., Rissland OS. *Systematic identification and characterization of eukaryotic and viral 2A peptide-bond-skipping sequences*. *Cell Reports* 2025. DOI: 10.1016/j.celrep.2025.115822
*These authors contributed equally.
9. **Morrison EJ.**, Shvartz D., Horton ER., Bekturova A., Koren I., Rissland OS. *UBR4 regulates a MetAP-dependent N-degron pathway*. Under review. DOI: <https://doi.org/10.1101/2024.10.03.616566>
10. **Morrison EJ.**, and Rissland OS. *The generation and consequences of N-terminal proteoform diversity*. *Cell Reports* 2025. DOI: 10.1016/j.celrep.2025.116275

Abstracts and poster presentations

1. Shields EJ., **Morrison EJ.**, Filbin-Wong ME. *Determination of Viral Cap-Independent Translation Enhancer RNA structure using Computer Modeling and Chemical Modification*. Presented at the Metropolitan State University of Denver Undergraduate Research Conference 2016. 22 April, Denver, Colorado.
2. **Morrison EJ.**, Shields EJ., Filbin-Wong ME. *Determination of Higher Order Structures in the Cap-Independent Translation Enhancer of the Black Currant Reversion Virus 2*. Presented at the Metropolitan State University of Denver Undergraduate Research Conference 2016. 22 April, Denver, Colorado.
3. **Morrison E.**, Leti F., Gerhard GS., Still C., and DiStefano JK. *Hepatic Expression of Chemokine Ligand 20 (CCL20) in Nonalcoholic Steatohepatitis is Modulated by miR-590-5p*. Presented at the Center for Genes, Environment and Health 2017. 27 February, Denver, Colorado.
4. **Morrison EJ.**, Rissland OS. *The N-terminus regulates gene expression in humans*. Presented at the Annual Gilliam Fellows Meeting 2021. 22 September, virtual conference.
5. Abudushalamu A., **Morrison EJ.**, Rissland OS. *METAP2 expression is regulated by translation elongation*. Presented at the SACNAS NDiSTEM Conference. 2022. 27. October, San Juan, Puerto Rico.

6. **Morrison EJ.**, Horton E., Rissland OS. Characterization of a novel N-terminal degron in human cells. Presented at the Howard Hughes Medical Institute December Meeting. 2022. 14 December, Chevy Chase, Maryland.
7. **Morrison EJ.**, Horton E., Rissland OS. UBR4 mediates the quality control of pen-ultimate Arg/N-recognins in mammals. Presented at the Howard Hughes Medical Institute Annual Gilliam Fellow Meeting. 2023. 31 March, Chevy Chase, Maryland.
8. **Morrison EJ.**, Horton E., Rissland OS. Dynamic regulation of tertiary N-degron motifs in humans. Presented at RNA Society 2023. 03 June, Singapore, Singapore.
9. **Morrison EJ.**, Horton E., Rissland OS. Tertiary lysines and arginines destabilize proteins in human cells. Presented at the Howard Hughes Medical Institute Annual Gilliam Fellow Meeting. 2023. 28 September, Chevy Chase, Maryland.
10. **Morrison EJ.**, Horton E., Rissland OS. Methionine cleavage by METAP2 produces primary N-degrons. Presented at the Howard Hughes Medical Institute December Meeting. 2023. 13 December, Chevy Chase, Maryland.
11. **Morrison EJ.**, Horton E., Rissland OS. UBR4 regulates a MetAP2-dependent Arg/N-degron pathway. Presented at the Howard Hughes Medical Institute December Meeting. 2024. 18 September, Chevy Chase, Maryland

Oral presentations

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| 05/2014 | Openness and closedness: metaphors in discourse
Undergraduate English Conference, Division of Linguistics
SpringHill Suites, Denver, CO |
| 05/2016 | Higher order structures in the cap-independent translation enhancer of the blackcurrant virus 2
Metropolitan State University of Denver, Department of Chemistry Speaker Series |
| 04/2021 | Primary protein sequence regulates gene expression
Metropolitan State University of Denver, Department of Chemistry Speaker Series
Invited speaker |
| 10/2023 | Tying up loose ends: Characterization of a novel N-degron pathway in humans
*Awarded Outstanding Research Presentation in Cell/Molecular Biology and Developmental Biology
SACNAS NDiSTEM 2023
Oregon Convention Center, Portland Oregon |
| 05/2024 | UBR4 regulates a MetAP2-dependent Arg/N-degron pathway
2024 SEP Rising Stars in Biomedical Sciences
Sanford Burnham Prebys, La Jolla, California
Invited Speaker |

Lectures in service of education and outreach

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| 02/2019 | From rAGs to riTChes: on becoming an RNA biochemist
DSST: Conservatory Green High School; Invited speaker. |
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03/2020	Where does it all go? Metabolism and the chemistry of life Aurora Science and Technology Middle School; Invited speaker.
08/2021	Writing a message: How the beginning of the ORF dictates gene expression Science Gong Show; Invited Speaker
05/2022	Those we render invisible: racialized gendered reductionism from an empirical point of view WiSTEM Annual Symposium 2022 University of Colorado-Denver, Anschutz Medical Campus

Honors, special recognitions, and awards

1999	Denver Mile High Scholar Samuels Elementary School
2000	Samuel's Elementary School Super Citizen Samuels Elementary School
2013	Provost's Honor Roll Metropolitan State University of Denver
2013	Metro Scholar Need Award Metropolitan State University of Denver \$1250.00
2014	Metro Scholar Need Award Metropolitan State University of Denver \$800.00
2016	LAS Student Research Grant Metropolitan State University of Denver \$500.00
2016	2016 Outstanding Student of the Year Nominee Metropolitan State University of Denver, College of Letters, Arts and Sciences
2018	RNA Bioscience Initiative Academic Enrichment Award University of Colorado-Denver, Anschutz Medical Campus \$5000.00
2020	Molecular Biology Predoctoral NRSA T32 Institutional Training Fellowship University of Colorado-Denver, Anschutz Medical Campus \$24,816.00
2021 – 2022	National Science Foundation Graduate Research Fellowship University of Colorado-Denver, Anschutz Medical Campus \$46,000.00 per year

- 2022 – Howard Hughes Medical Institute Gilliam Fellowship
University of Colorado-Denver, Anschutz Medical Campus
\$55,000.00 per year
- 2023 SACNAS NDiSTEM Travel Scholarship
University of Colorado-Denver, Anschutz Medical Campus
- 2023 Outstanding Research Presentation in Cell/Molecular Biology and Developmental Biology
NDiSTEM SACNAS 2023, Portland, Oregon
- 2024 Sanford Burnham Prebys 2024 Rising Stars in Biomedical Sciences
Stanford Burnham Prebys, La Jolla, California
- 2025 AAAS/*Science* Program for Excellence in Science
Nominated by Dr. Jennifer Richer, Dean of CU Anschutz Graduate School
University of Colorado-Denver, Anschutz Medical Campus

Professional affiliations

- 2019 – Present Society for the Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)
National Student Member
- 2019 – Present American Society for Biochemistry and Molecular Biology (ASBMB)
University of Colorado-Denver, Anschutz Medical Campus
Student Member
- 2018 – Present RNA Society
University of Colorado-Denver, Anschutz Medical Campus
Student Member
- 2014 – Present Golden Key International Honor Society
Metropolitan State University of Denver
Student Member

Committee and service responsibilities

- 2024 – Black Graduate Student Association
University of Colorado-Denver, Anschutz Medical Campus
Student Member
- 2022 – 2025 RNA Bioscience Initiative Summer Internship Program Admission Committee
University of Colorado-Denver, Anschutz Medical Campus
Reviewer
- 2021 – 2022 Women in Science, Technology, Engineering and Mathematics (WiSTEM)
University of Colorado-Denver, Anschutz Medical Campus
Ally Coordinator

2021 – 2023	Molecular Biology Graduate Advisory Committee University of Colorado-Denver, Anschutz Medical Campus Student Representative
2020 – 2023	Diversity, Equity and Anti-Racism (DEA-R) Recruitment Committee Biochemistry and Molecular Genetics University of Colorado-Denver, Anschutz Medical Campus Co-chair
2020 – 2022	Seminar Committee Biochemistry and Molecular Genetics University of Colorado-Denver, Anschutz Medical Campus Student Member
2020 – 2022	National Science Foundation Graduate Research Fellowship Review Committee University of Colorado-Denver, Anschutz Medical Campus Reviewer
2019 – 2020	Molecular Biology T32 Round Table Planning Committee University of Colorado-Denver, Anschutz Medical Campus Student Member
2019 – 2025	Society for the Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS) Local Student Member
2014 – 2016	Louis Stokes Colorado Alliance for Minority Participation Metropolitan State University of Denver Student Member

Teaching record

Undergraduate

<u>CHE 4350</u> 01/2015 – 05/2015	Biochemistry laboratory Teaching Assistant Metropolitan State University of Denver
<u>CHE 3000</u> 08/2015 – 12/2015	Analytical chemistry Supplemental Instructor Metropolitan State University of Denver
<u>CHE 4310</u> 01/2016 – 05/2016	Biochemistry I Supplemental Instructor Metropolitan State University of Denver

Graduate

<u>MOLB 7950</u>	Bioinformatics and statistics
08/2022 – 12/2022	Teaching Assistant University of Colorado-Denver, Anschutz Medical Campus
<u>IDPT 7806</u>	Foundations in Biomedical Sciences
08/2022 – 12/2022	Teaching Assistant University of Colorado-Denver, Anschutz Medical Campus

Mentoring and training***Undergraduate***

12/2015 – 09/2022	Laura Baquero (student: Biology major) Metropolitan State University of Denver Currently Doctoral Candidate in the Molecular and Cellular Biology Program at University of Washington
06/2022 – 08/2022	Adina Abudushalamu (student: Biochemistry & Computer Science major) University of Colorado Denver, Anschutz Medical Campus Currently Graduate Student at the University of California, Davis
06/2023 – 08/2023	Emily Allen (student: Biology major) University of Colorado Denver, Anschutz Medical Campus Currently Graduate Student at University of Michigan
09/2023 – 05/2024	Edgar Provencio (student: Biology Major) University of Colorado-Denver, Anschutz Medical Campus Currently student in the Biology Program at Metropolitan State University of Denver
06/2024 – 06/2025	Tan Hyunh (student: Biology Major) University of Colorado-Denver, Anschutz Medical Campus Currently student in the Biology Program at Metropolitan State University of Denver

Professional Research Assistants and graduate students

02/2022 – 06/2022	Emma Horton (Professional Research Assistant) University of Colorado Denver, Anschutz Medical Campus Currently Research Assistant at the Allen Institute, Seattle Washington
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Community service

12/2012– 03/2014	Mentor and Volunteer Coordinator Aurora Youth Options, Aurora, Colorado
02/2019 – 12/2021	Sexual Assault Victims Advocate The Blue Bench, Denver, Colorado

2019 – 2020	RNA Bioscience Initiative Summer Internship Program Group Leader University of Colorado-Denver, Anschutz Medical Campus
2019	Neuroanatomy Teaching Assistant, SACNAS Summer Academy University of Colorado-Denver, Anschutz Medical Campus
09/2020 – 03/2023	Bryant Webster STEM Scholar Mentor University of Colorado-Denver, Anschutz Medical Campus
06/2023 – 09/2023	CC2STEM Mentor University of Colorado-Denver, Anschutz Medical Campus